Decision trees
Bagging

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Recap

Lecture 4:
SVM, PCA
kNN indexes

1. Support Vector Machine (SVM)
   - Hinge loss
   - Kernel trick
2. Dimensionality reduction and PCA
   - Problem statement
   - Singular Value Decomposition
   - Eckart–Young theorem
   - Equivalent definitions
   - Data normalization
3. k Nearest Neighbors
   - kNN indexes
   - HNSW

Outline

1. Intuition
2. Construction procedure
3. Information criteria
4. Special highlights
    - Dealing with missing data
    - Binarization
    - Decision tree as linear model
    - Standards
    - Hyperparameters
5. Bootstrap and Bagging
6. Random Forest

Intuition

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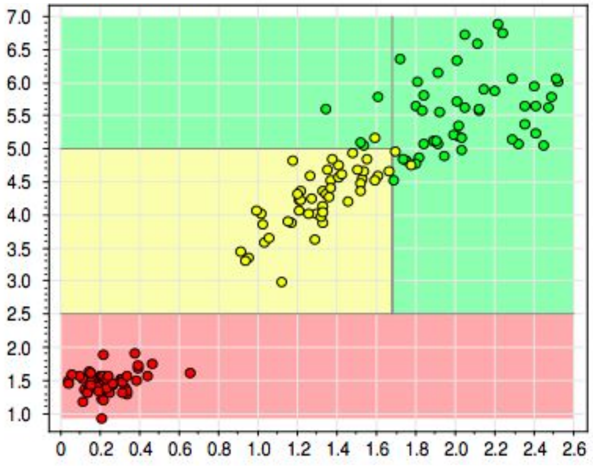
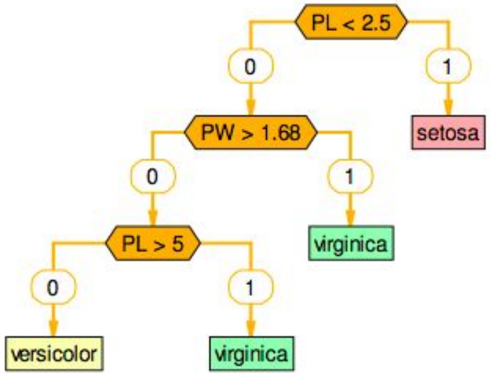
01

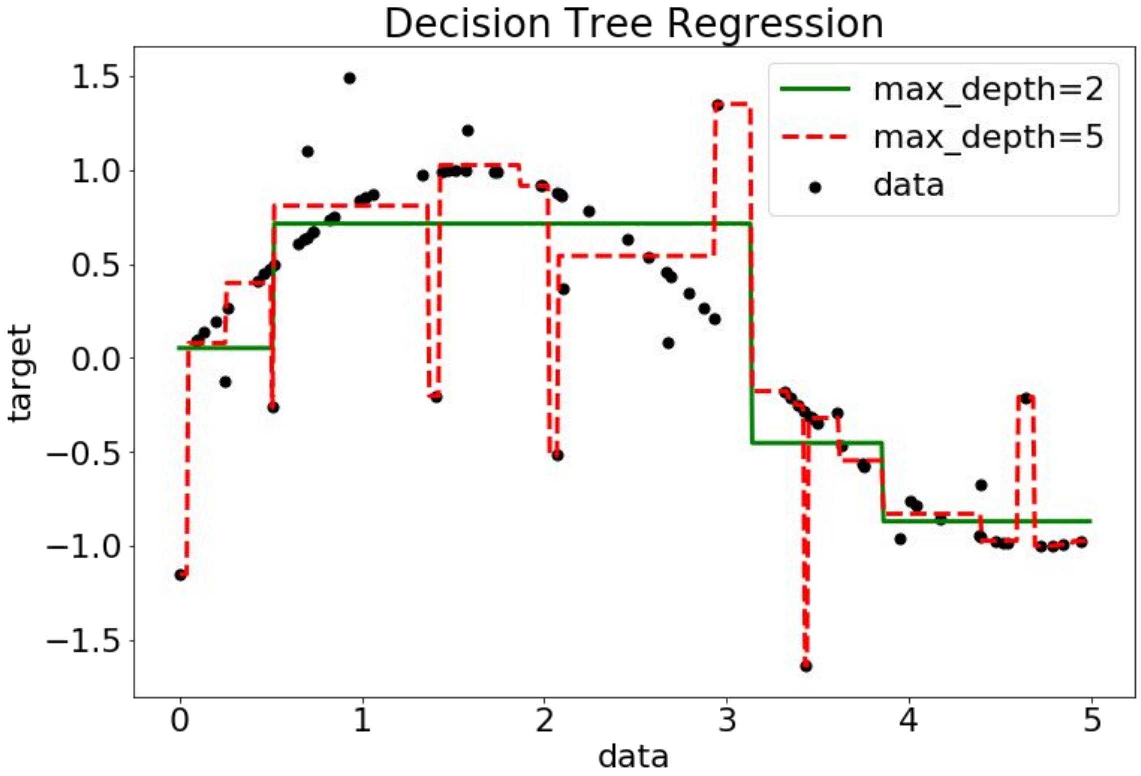
Decision tree for Fisher’s Iris dataset





setosa	$r_1(x) = [PL \leqslant 2.5]$
virginica	$r_2(x) = [PL > 2.5] \wedge [PW > 1.68]$
virginica	$r_3(x) = [PL > 5] \wedge [PW \leqslant 1.68]$
versicolor	$r_4(x) = [PL > 2.5] \wedge [PL \leqslant 5] \wedge [PW < 1.68]$

Decision tree in regression

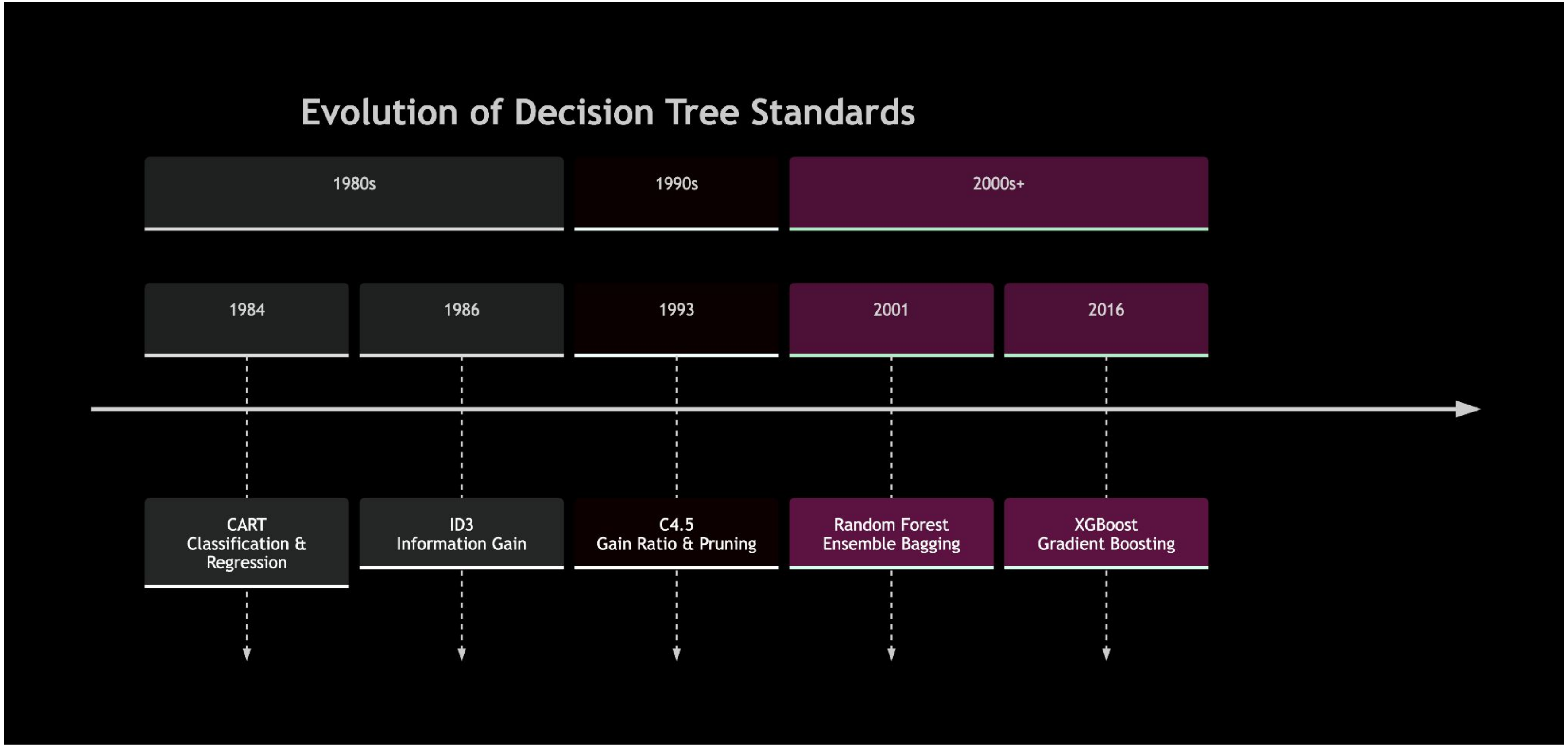


Green - decision tree of depth 2

Red - decision tree of depth 5

Every leaf corresponds to some constant.

Timeline of tree algorithms



Standards

- Iterative Dichotomiser 3 (ID3) (1986)
  - Entropy criteria; Stops when no more gain available
- C4.5 (1993)
  - Normalised entropy criteria; Stops depending on leaf size; Incorporates pruning
- ID5 (1994)
  - Introduces iterative learning enabling tree update after training
- CART (1984)
  - Gini criteria; Cost-complexity Pruning; Surrogate predicates for missing data;
  - Origin for most current implementations (sklearn, boosting libs)
- etc.

Read more

Construction procedure

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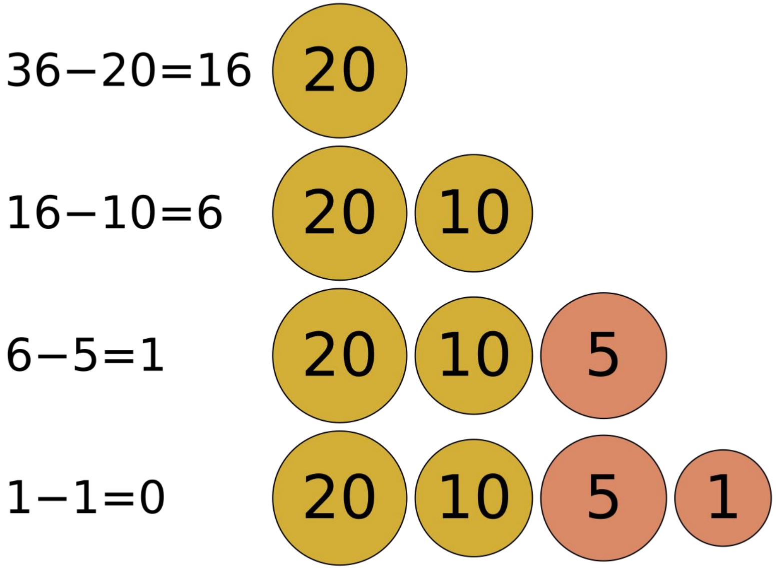
02

Greedy algorithms

A greedy algorithm is any algorithm that follows the problem-solving heuristic of making the locally optimal choice at each stage.

In many problems, a greedy strategy does not produce an optimal solution, but a greedy heuristic can yield locally optimal solutions that approximate a globally optimal solution in a reasonable amount of time.



Constructing decision trees

threshold value

j feature

$$x^{(j)} < t$$

True

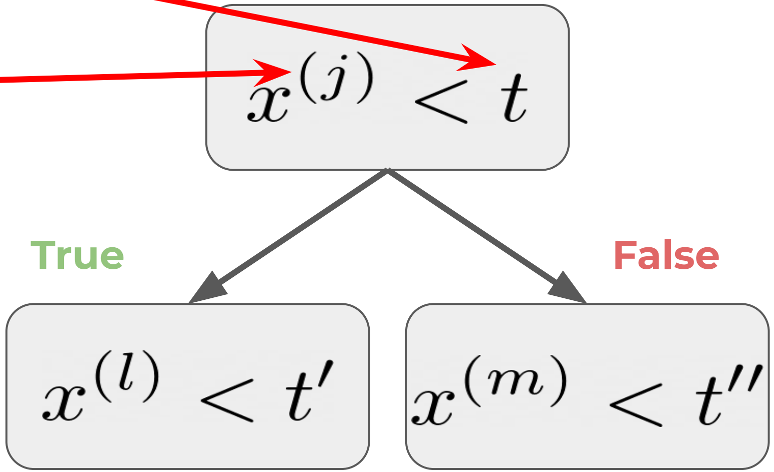
False

1. Make a split

Constructing decision trees

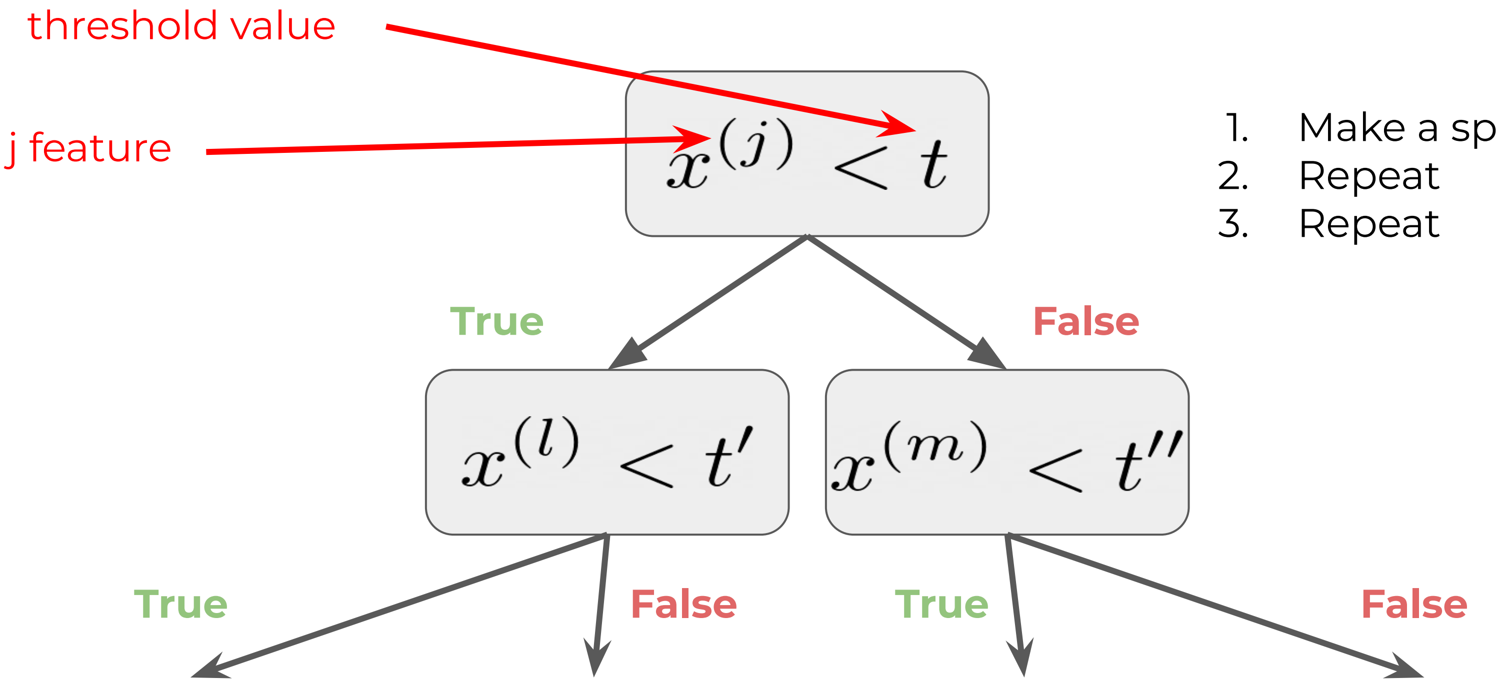
threshold value

j feature



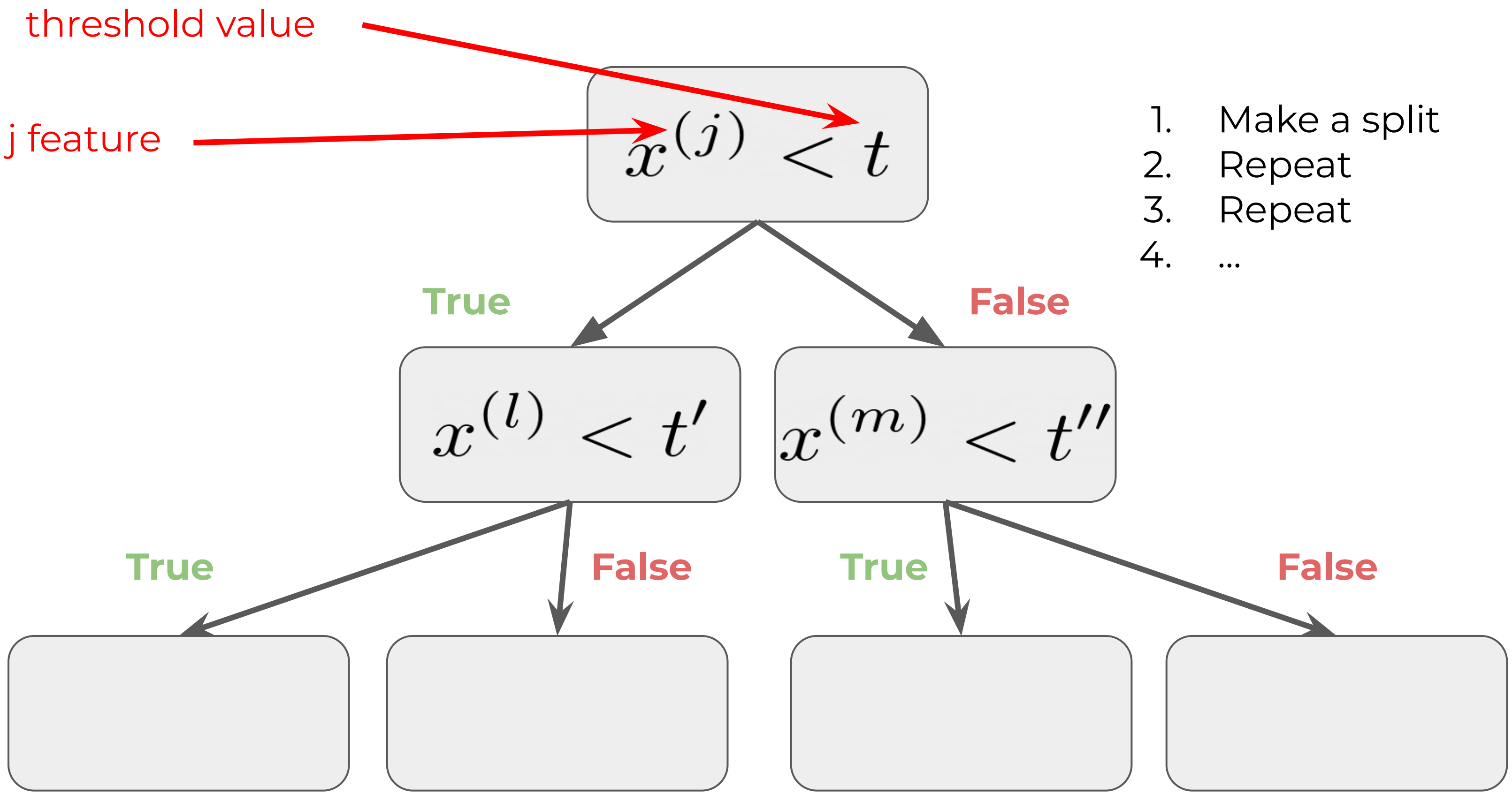
1. Make a split
2. Repeat

Constructing decision trees



1. Make a split
2. Repeat
3. Repeat

Constructing decision trees



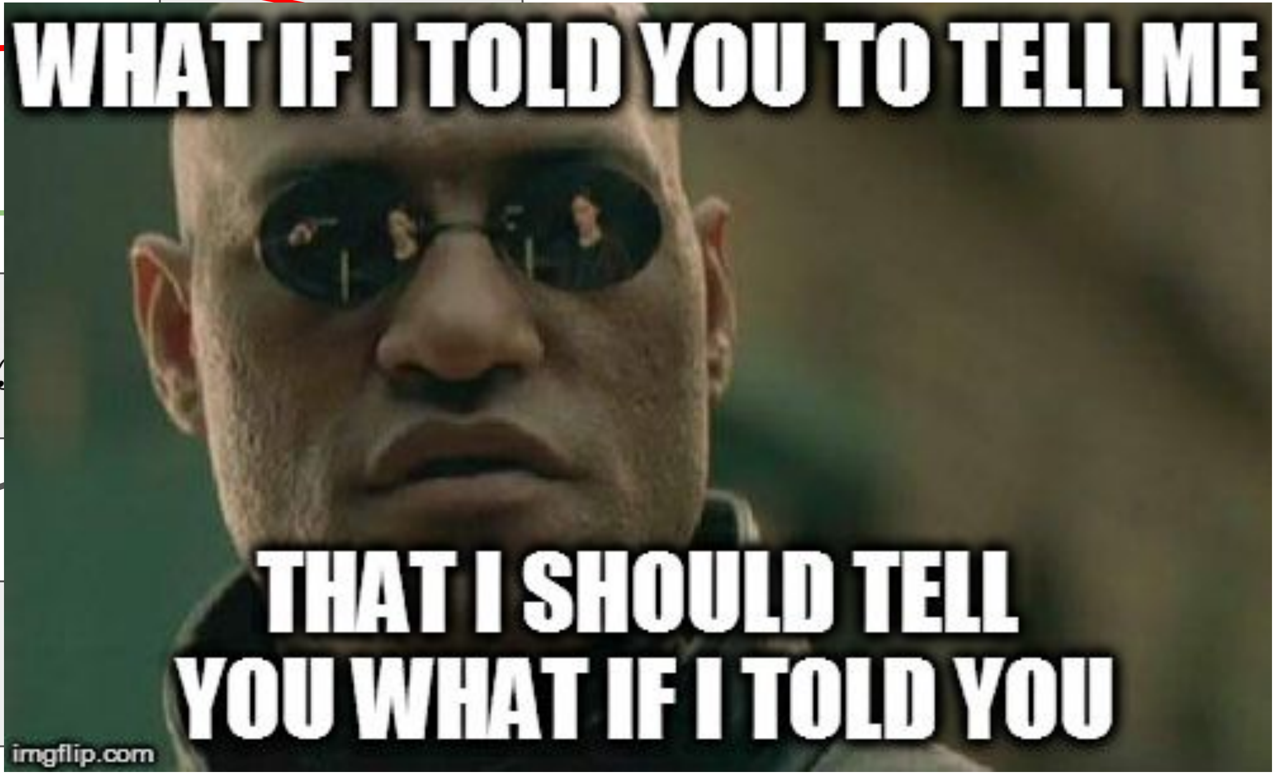
1. Make a split
2. Repeat
3. Repeat
4. …

threshold value

j feature

T

True



How to answer in leaf?

Classification:

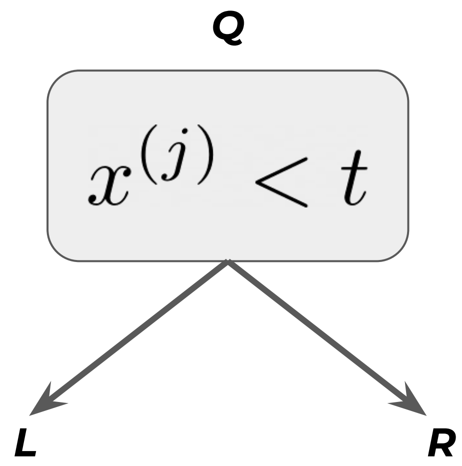
- most popular
- sample with frequencies of classes *(is it good enough?)*

Regression:

Depends on loss function!

- for MSE
  - average in node
- for MAE
  - median in node

How to split data properly?



We can not use gradient this time because solution set is discrete.

So let's apply discrete optimization!

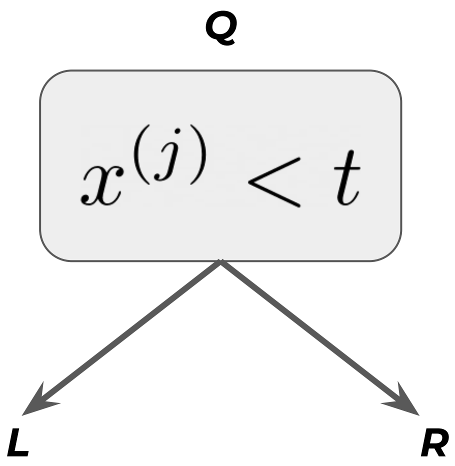
$$\frac{|L|}{|Q|}H(L) + \frac{|R|}{|Q|}H(R) \longrightarrow \min_{j,t}$$

How to choose particular split?

Brute force algorithm will take too much time.

Random splits are chosen and compared.

How to split data properly?



What is H?

$$\frac{|L|}{|Q|}H(L) + \frac{|R|}{|Q|}H(R) \longrightarrow \min_{j,t}$$

Information criteria

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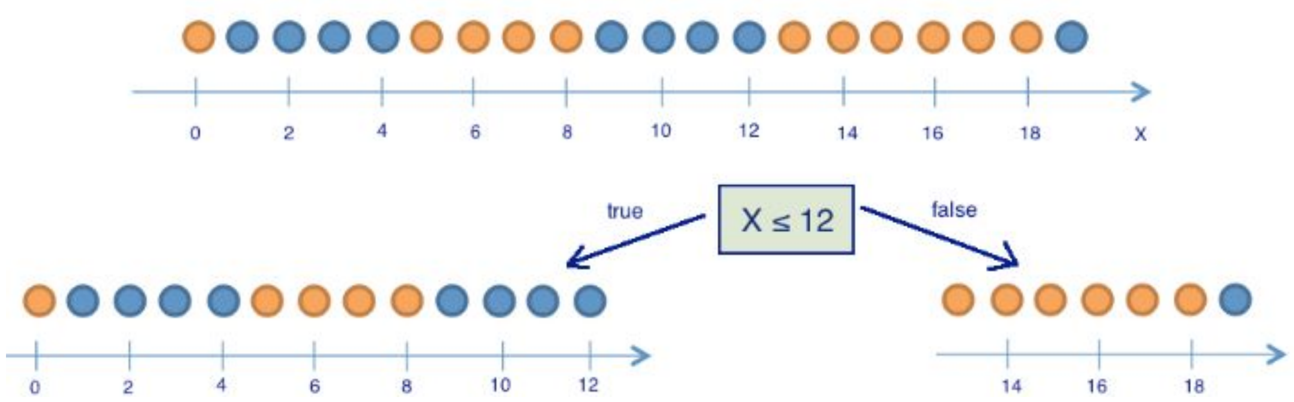
03

Information criteria

H(R) is measure of “heterogeneity” of our data.

Consider binary classification problem:



Binary information criteria

H(R) is measure of “heterogeneity” of our data.

Consider binary classification problem:

0. Misclassification criteria: $$H(R) = 1 - \max\{p_0, p_1\}$$

1. Entropy criteria: $$H(R) = -p_0 \log p_0 - p_1 \log p_1$$

2. Gini impurity: $$H(R) = 1 - p_0^2 - p_1^2 = 2p_0p_1$$

Multiclass information criteria

H(R) is measure of “heterogeneity” of our data.

Consider multiclass classification problem:

0. Misclassification criteria:

$$H(R) = 1 - \max_k \{p_k\}$$

1. Entropy criteria:

$$H(R) = -\sum_{k=0}^{K} p_k \log p_k$$

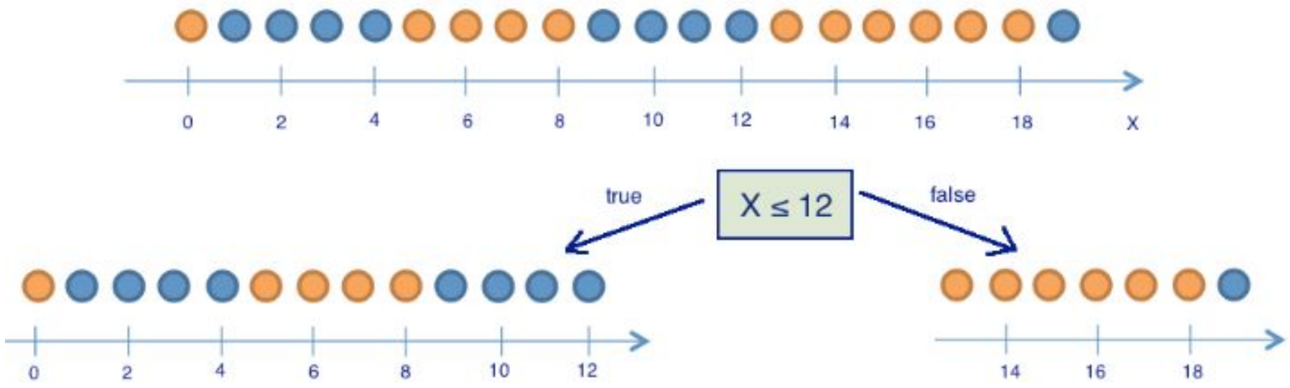
2. Gini impurity:

$$H(R) = 1 - \sum_k (p_k)^2$$

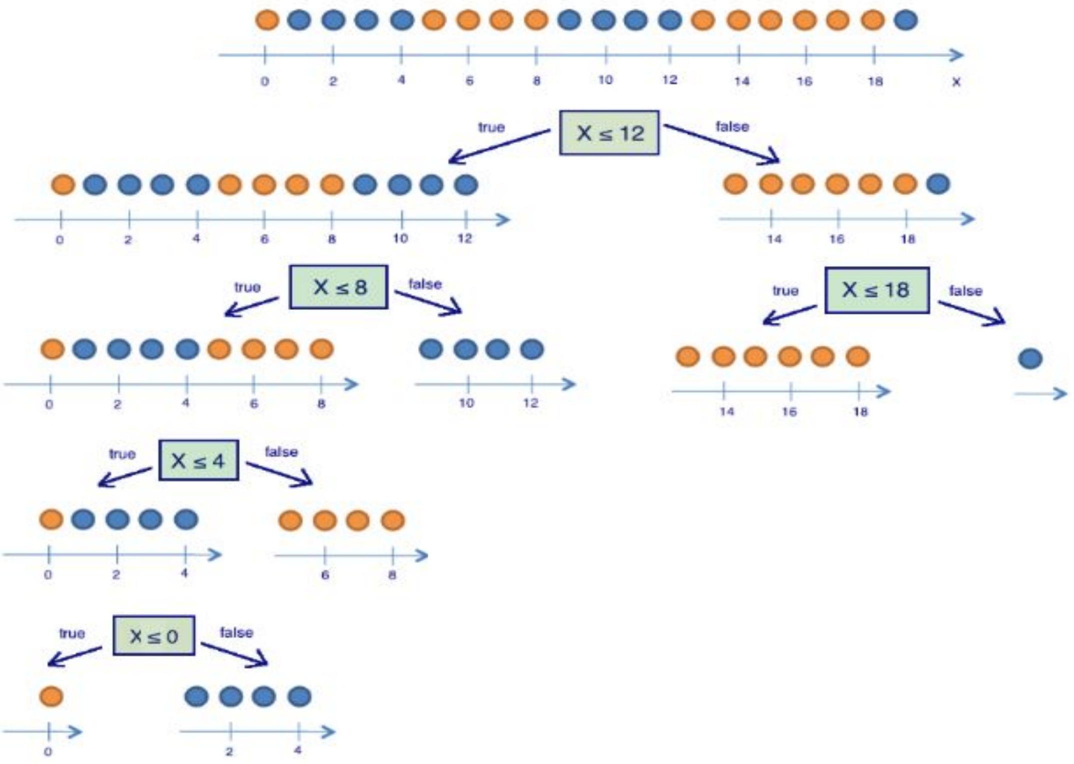
Information criteria

H(R) is measure of “heterogeneity” of our data.

Consider binary classification problem:



Entropy



$$S = -M \sum_{k=0}^{K} p_k \log p_k$$

In binary case N = 2

$$S = -p_+ \log_2 p_+ - p_- \log_2 p_- = -p_+ \log_2 p_+ - (1 - p_+) \log_2 (1 - p_+)$$

Gini impurity

$$G = 1 - \sum_k (p_k)^2$$

In binary case N = 2

$$G = 1 - p_+^2 - p_-^2 = 1 - p_+^2 - (1 - p_+)^2 = 2p_+(1 - p_+)$$

Gini impurity

$$G = 1 - \sum_k (p_k)^2$$

1. missclass prob
2. Var of Bernoulli
3. Taylor of Ent

$H = \sum p \log p \approx \sum p(1-p)$

In binary case N = 2

Taylor: $\log(p) \underset{p \to 1}{\approx} 1 - p$

$$G = 1 - p_+^2 - p_-^2 = 1 - p_+^2 - (1 - p_+)^2 = 2p_+(1 - p_+)$$

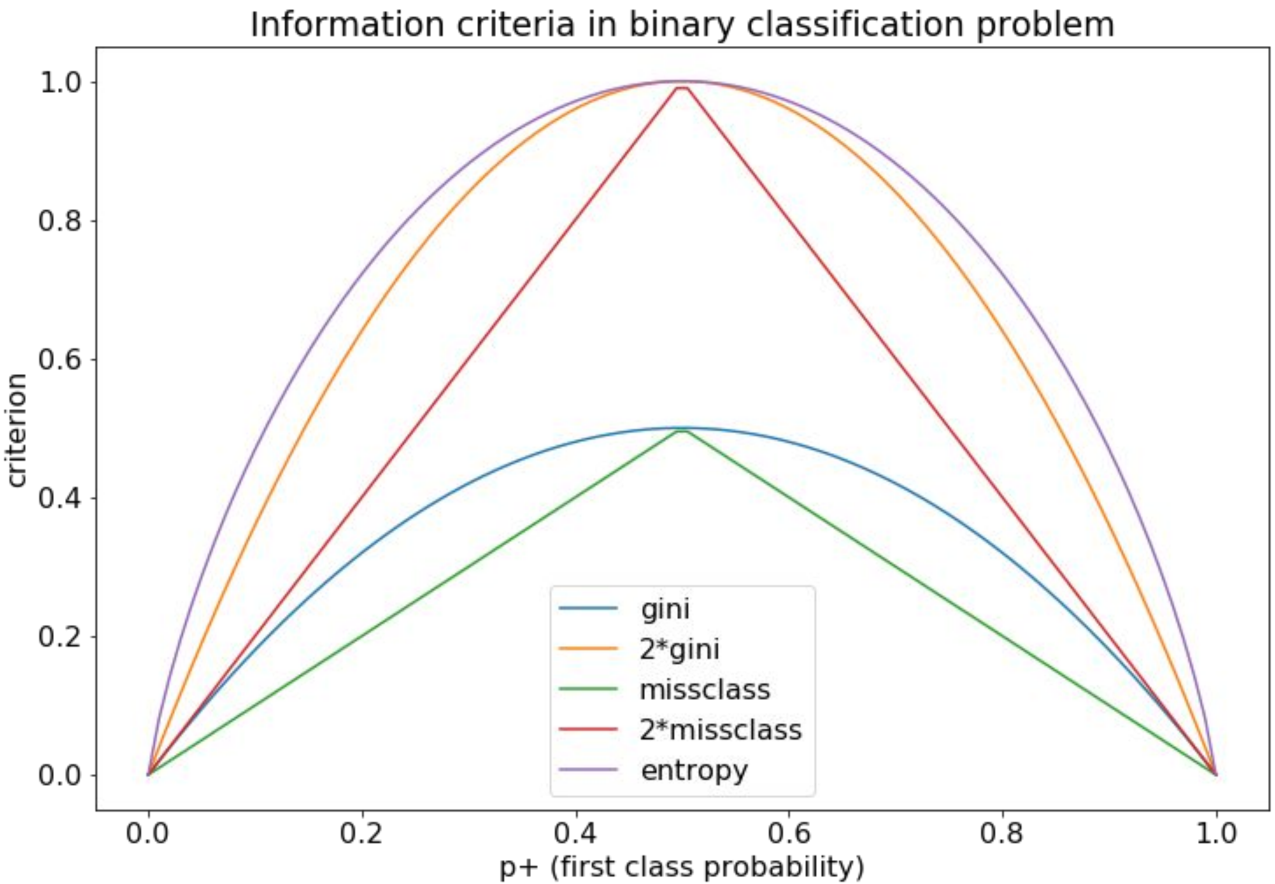
$p : 1$

$1-p : 0$

$\mathbb{E}(X) = p - (1-p) = p$

$Var(X) = \mathbb{E}x^2 - (\mathbb{E}x)^2 = p - p^2 = p(1-p)$

Comparison


Information criteria in binary classification problem
1.0
0.8
0.6
0.4
0.2
0.0
criterion
0.0
0.2
0.4
0.6
0.8
1.0
p+ (first class probability)
gini
2*gini
missclass
2*missclass
entropy

Regression case

H(R) is measure of “heterogeneity” of our data.

Consider regression problem:

1. Mean squared error

$$H(R) = \min_c \frac{1}{|R|} \sum_{(x_i, y_i) \in R} (y_i - c)^2$$

What is the constant?

$$c^* = \frac{1}{|R|} \sum_{y_i \in R} y_i$$

Decision Trees as Linear models

Let J be the subspace of the original feature space, corresponding to the leaf of the tree.

Prediction takes form

$$\hat{y} = \sum_j w_j [x \in J_j]$$

Decision Trees as Linear models

Normalization in trees

Trees are not affected by denormalized data because they choose volume spaces regardless of the scale of both features and target

$$\hat{y} = \sum_j w_j [x \in J_j]$$

Parameters

Variable number of parameters!

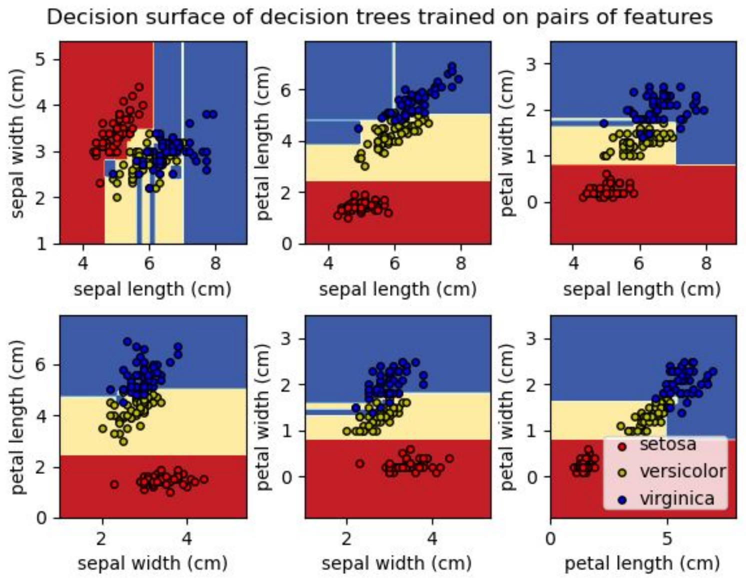
Hyperparameters

- max_depth: min 1
- min_samples_split: min 2
- min_samples_leaf: min 1
- min_impurity_decrease

Minor

- criterion:
  - gini, entropy for classification
  - MSE or MAE for regression
- splitter: best, random
- max_features: sqrt, log2

As of sklearn implementation



Such hparams could be named pre-pruning or inductive bias

Special highlights

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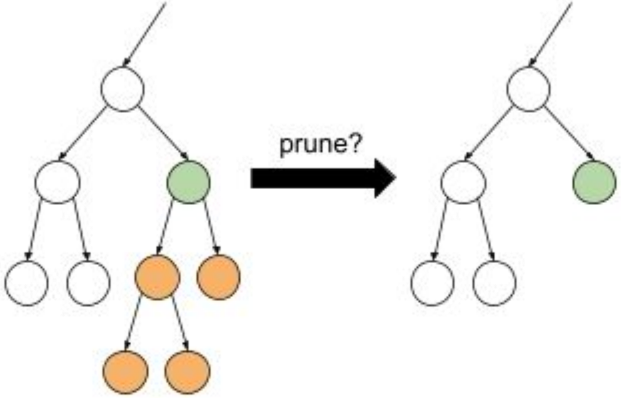
04

Pruning

Process of tree simplification after initial greedy algorithm to prevent overfitting

[video on pruning]



C4.5 pruning

Pessimistic Pruning

C4.5 uses a binomial confidence interval to estimate the "true" error rate. It assumes that the observed errors on the training set are just a sample and the real error could be higher.
It calculates an upper bound for the error rate of the subtree and the leaf node, assuming a certain confidence level (e.g., 25% is used as a default, which correlates roughly with a standard statistical confidence interval).

CART pruning

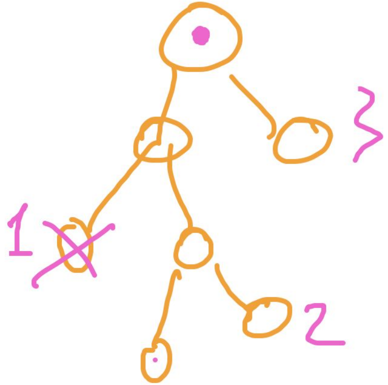
val (cross-val)

$\alpha_1 < \alpha_2 < \alpha_3 \dots$

Cost-Complexity based

R_α(T) = R(T) + α|T|

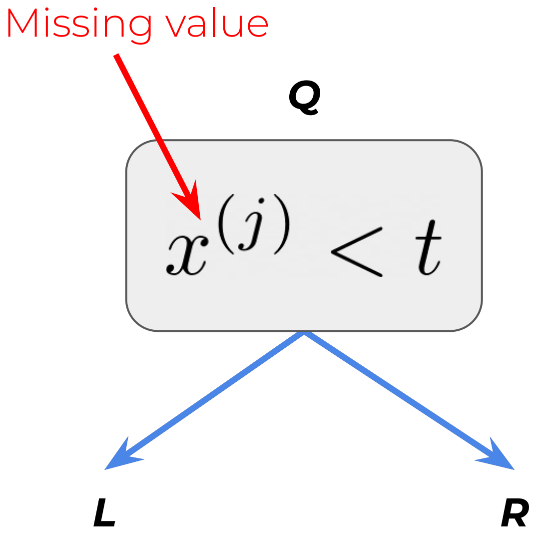
For each non-leaf node in the current tree, calculate how much the α would have to increase for pruning that node to become better than keeping its subtree. This is the "effective α" of the node. Sequence of increasing alphas generate sequence of trees.



Missing values in Decision Trees

In C4.5 if the value is missing, one might use both sub-trees and average their predictions.

But this will negatively affect model computational performance.



$$\hat{y} = \frac{|L|}{|Q|}\hat{y}_L + \frac{|R|}{|Q|}\hat{y}_R$$

Missing values in Catboost

Forbidden: Missing values are not supported, their presence is interpreted as an error

Min: Missing values are processed as the minimum value (less than all other values) for the feature. It is guaranteed that a split that separates missing values from all other values is considered when selecting trees.

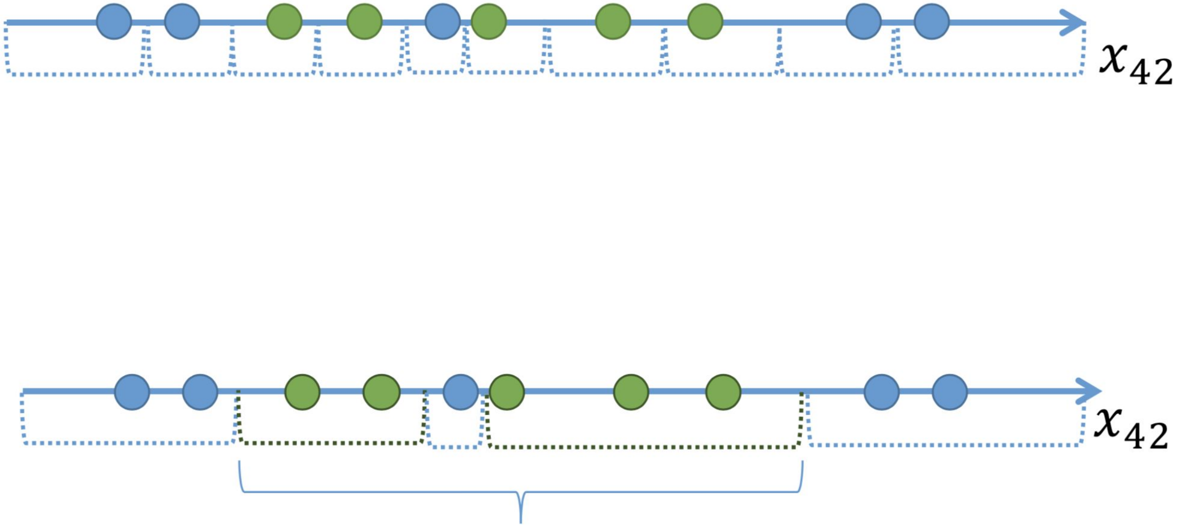
Max: Missing values are processed as the maximum value (greater than all other values) for the feature. It is guaranteed that a split that separates missing values from all other values is considered when selecting trees.

The **default** processing mode **is Min**

Documentation

Binarization

Idea: instead selecting one threshold define several for one feature.



e.g. [Border count hyperparameter]() in Catboost (defaults to 254)

Bootstrap and Bagging

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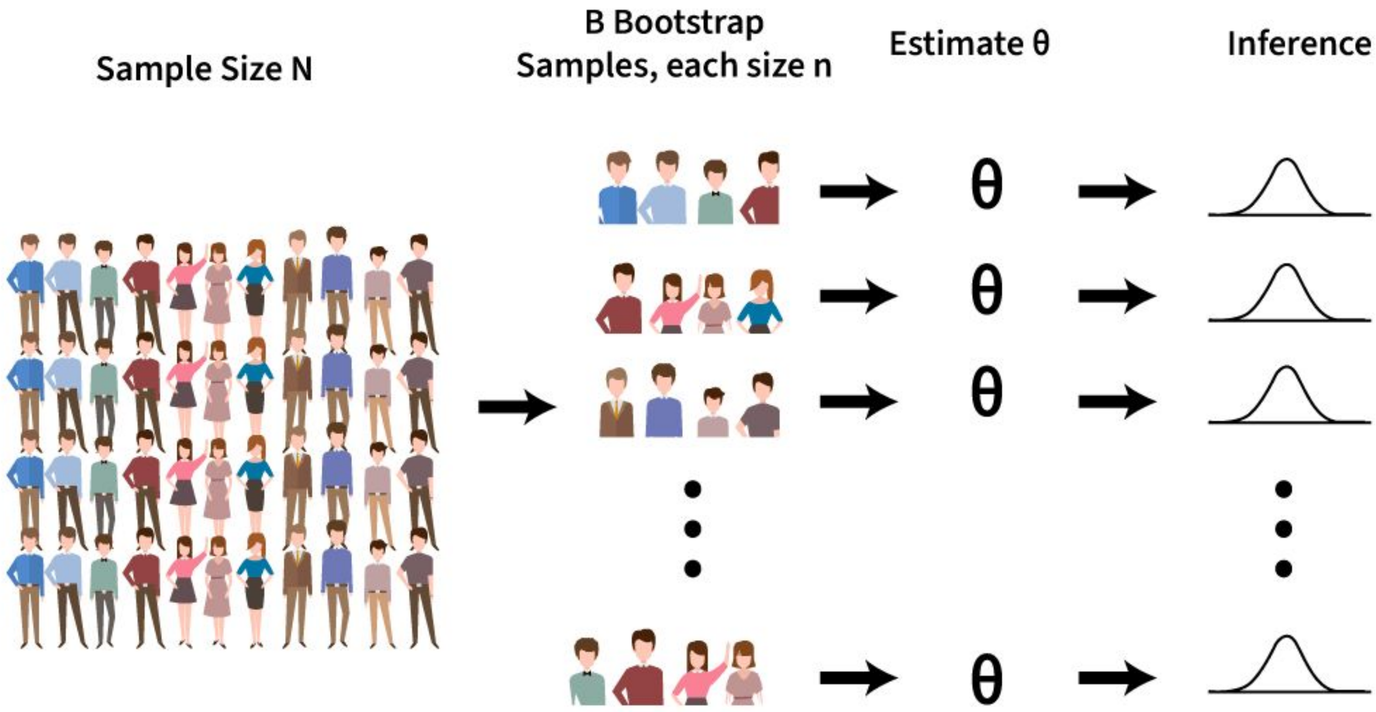
06

Bootstrap



https://www.youtube.com/watch?v=Xz0x-8-cgaQ

Bootstrap

Consider dataset X containing m objects.

Pick m objects with return from X and repeat in N times to get N datasets.

Error of model trained on Xj: $$\varepsilon_j(x) = b_j(x) - y(x), \qquad j = 1, \dots, N,$$

Then $$\mathbb{E}_x(b_j(x) - y(x))^2 = \mathbb{E}_x \varepsilon_j^2(x).$$

The mean error of N models: $$E_1 = \frac{1}{N} \sum_{j=1}^{N} \mathbb{E}_x \varepsilon_j^2(x).$$

Bootstrap

Consider the errors ~~unbiased and uncorrelated:~~

This is a lie

$$\mathbb{E}_x \varepsilon_j(x) = 0;$$

$$\mathbb{E}_x \varepsilon_i(x)\varepsilon_j(x) = 0, \quad i \neq j.$$

$$E_N = \mathbb{E}_x \left(\frac{1}{N} \sum_{j=1}^{n} b_j(x) - y(x) \right)^2 =$$

$$= \mathbb{E}_x \left(\frac{1}{N} \sum_{j=1}^{N} \varepsilon_j(x) \right)^2 =$$

$$= \frac{1}{N^2} \mathbb{E}_x \left(\sum_{j=1}^{N} \varepsilon_j^2(x) + \underbrace{\sum_{i \neq j} \varepsilon_i(x)\varepsilon_j(x)}_{=0} \right) =$$

$$\boxed{= \frac{1}{N} E_1.}$$

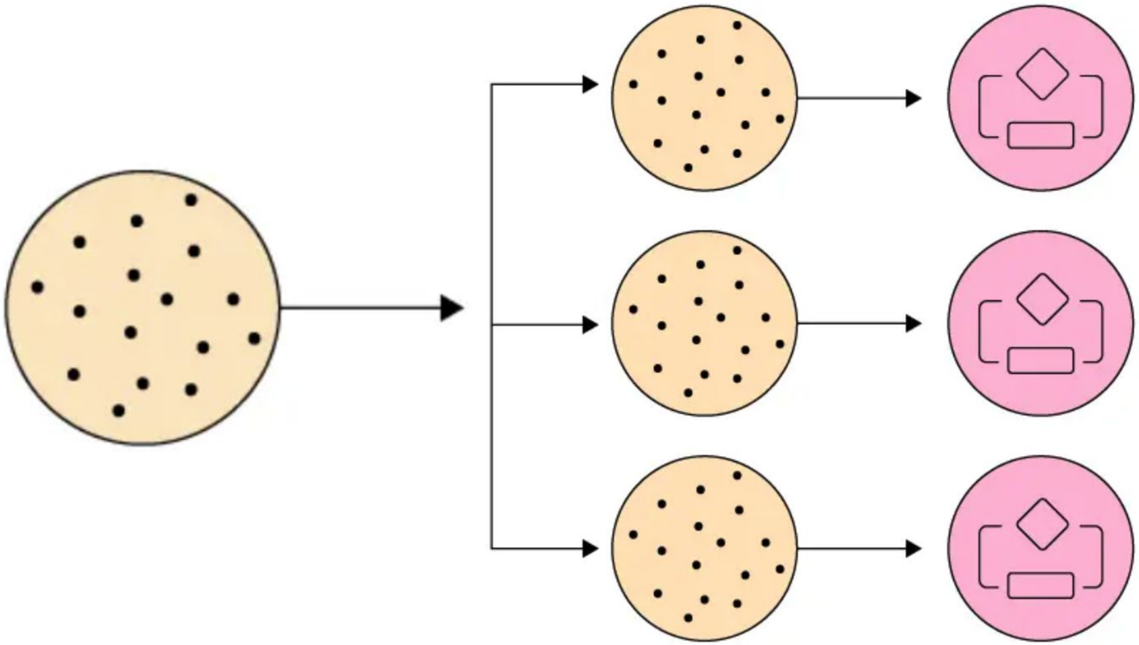
The final model averages all predictions:

$$a(x) = \frac{1}{N} \sum_{j=1}^{N} b_j(x).$$

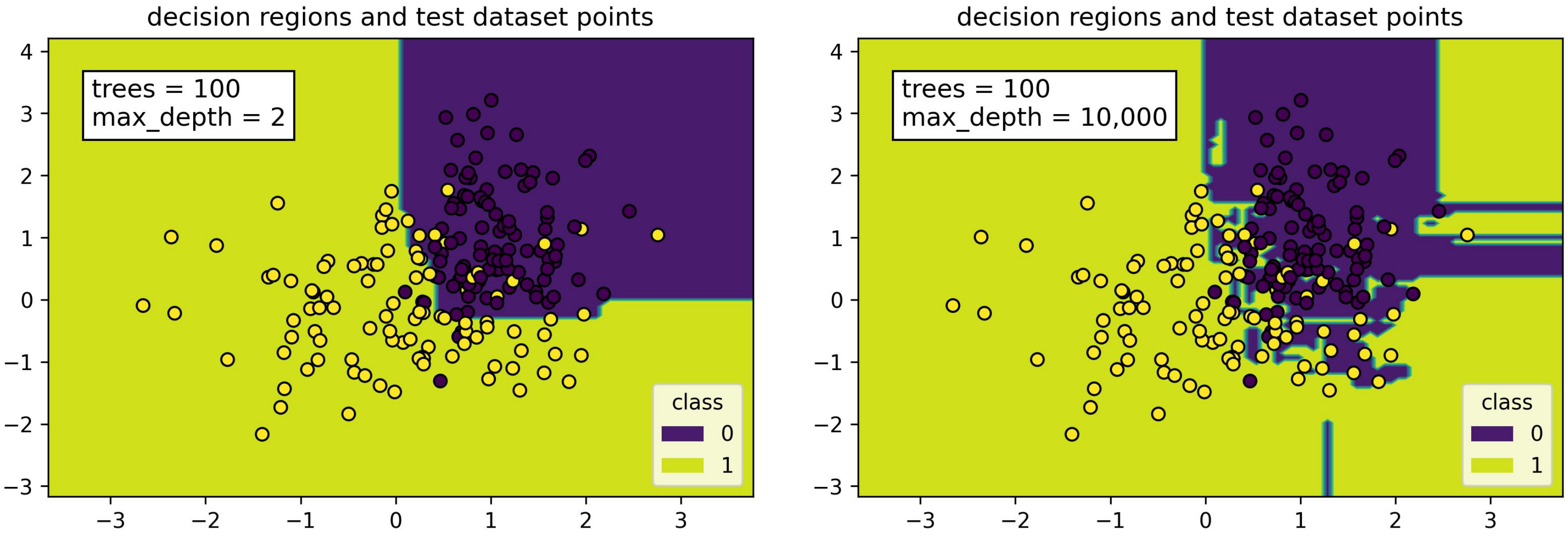
Error decreased by N times!

Bagging = Bootstrap aggregating

Decreases the variance if the basic algorithms are not correlated



Bagging overfitting



https://www.tommykhoo.com/machine-learning/rf_overfit/

Random Forest

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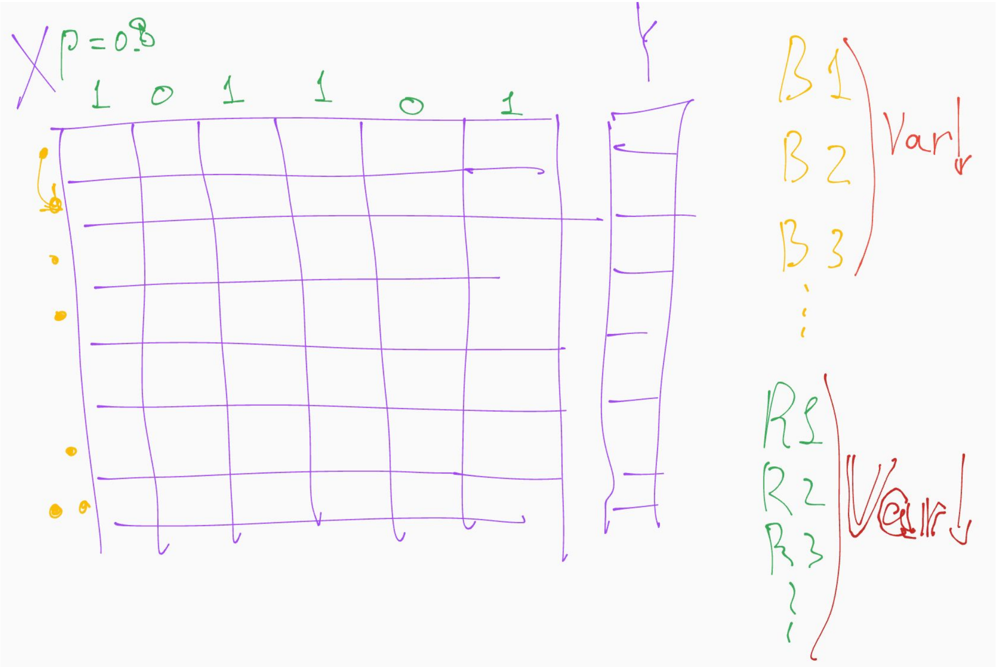
07

Random Subspace Method (RSM)

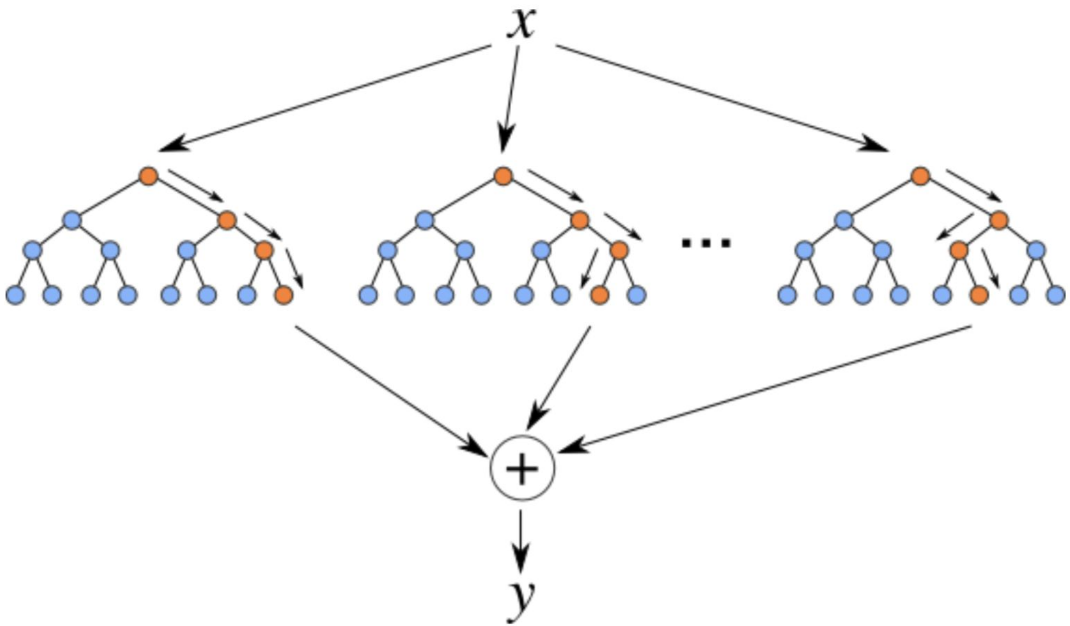
Same approach, but with features.

Just subsample of features for each bootstrapped dataset



Random Forest

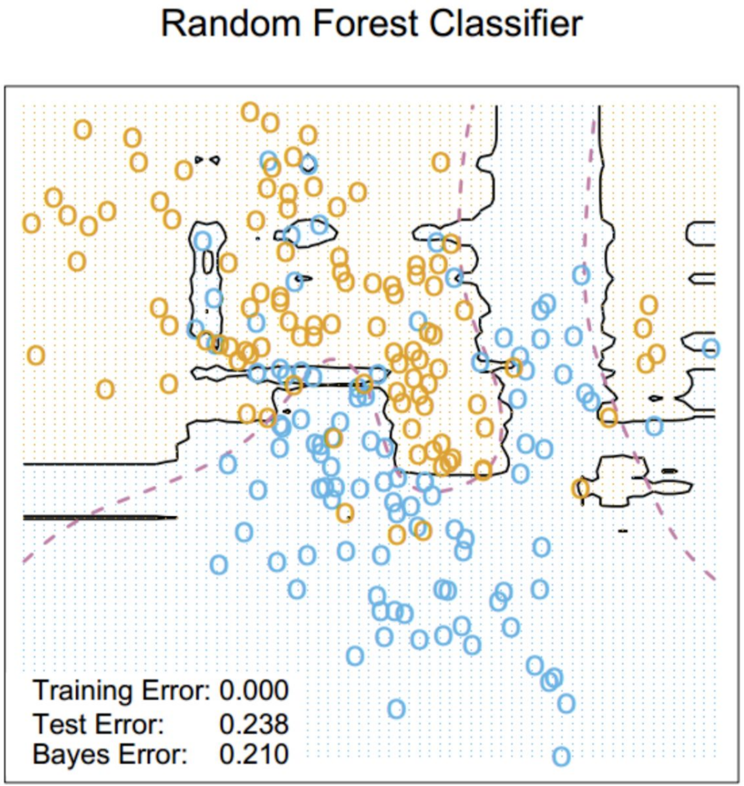
Bagging + RSM = Random Forest

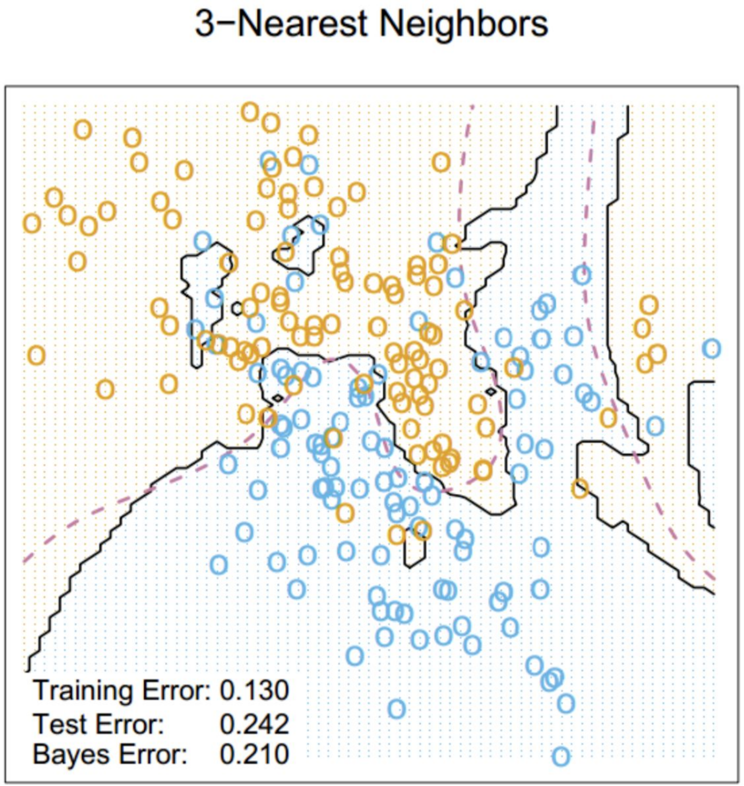


Random Forest summary

- Induces additional Variance in data to suppress it then with trees
- One of the greatest “universal” models
- There are some modifications
  - Extremely Randomized Trees
  - Isolation Forest

Comparison to kNN





Isolation forest

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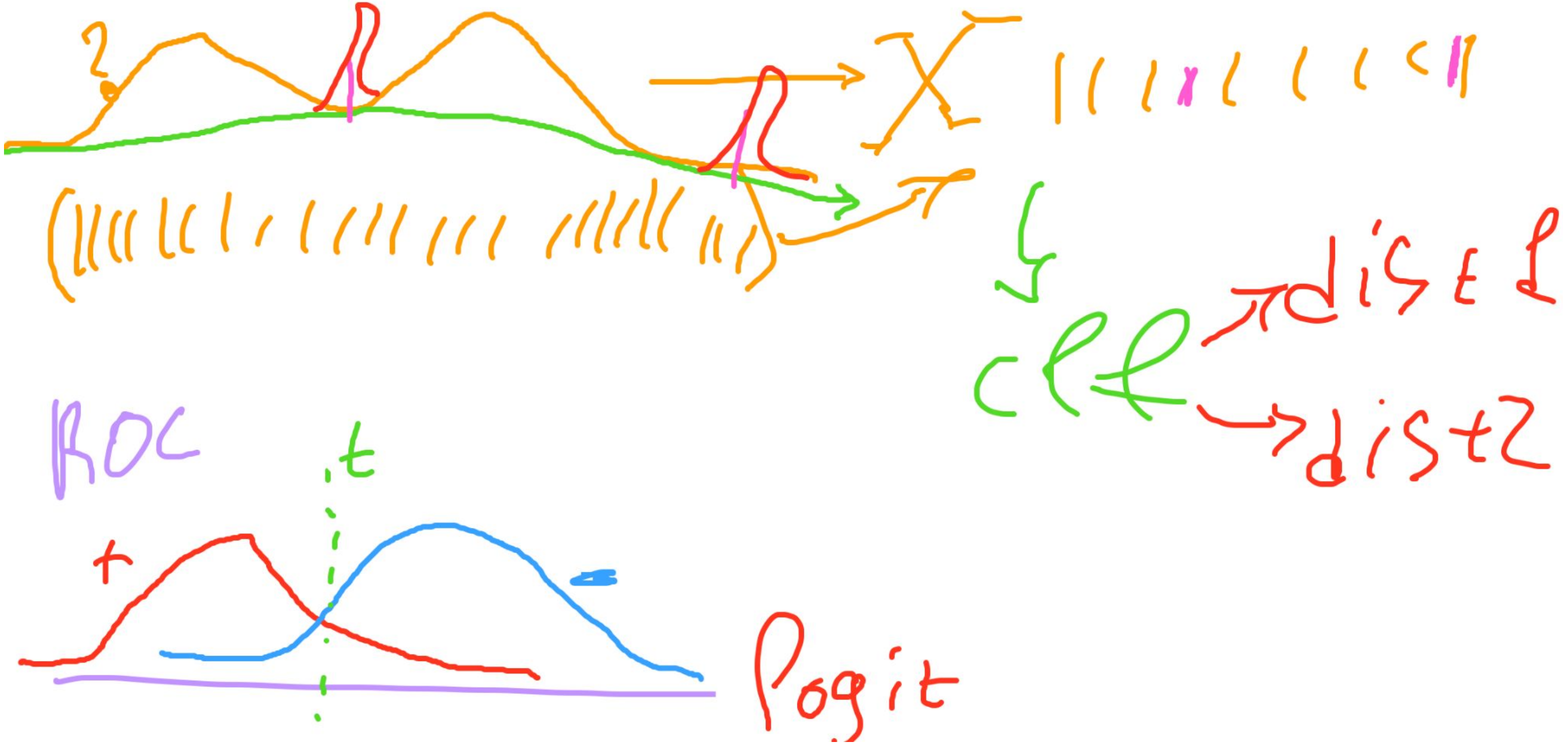
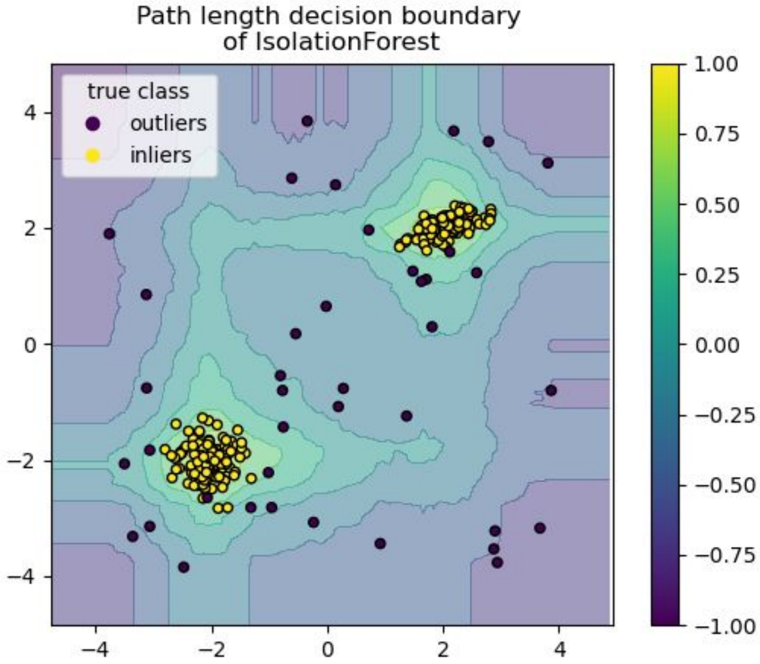
08

Anomaly detection

We have to distinguish out of distributions samples.



Method to search for anomalies



Method to search for anomalies

Isolation Forest ‘isolates’ observations by randomly selecting a feature and then randomly selecting a split value between the maximum and minimum values of the selected feature.

This path length, averaged over a forest of such random trees, is a measure of normality and our decision function.

Random partitioning produces noticeably shorter paths for anomalies. Hence, when a forest of random trees collectively produce shorter path lengths for particular samples, they are highly likely to be anomalies.

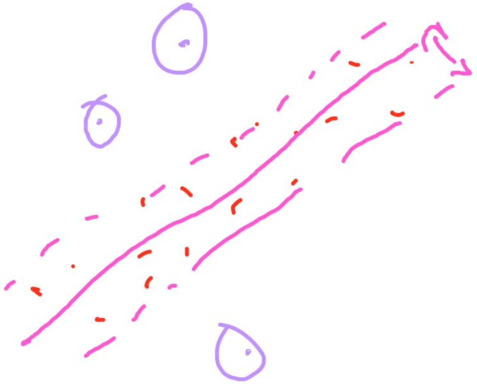
https://scikit-learn.org/stable/modules/outlier_detection.html#isolation-forest

https://alexanderdyakonov.wordpress.com/2017/04/19/%D0%BF%D0%BE%D0%B8%D1%81%D0%BA-%D0%B0%D0%BD%D0%BE%D0%BC%D0%B0%D0%BB%D0%B8%D0%B9-anomaly-detection/

One class SVM

https://scikit-learn.org/stable/modules/generated/sklearn.svm.OneClassSVM.html

One Class SVM



Revise

1. Intuition
2. Construction procedure
3. Information criteria
4. Decision trees special highlights
    - Decision tree as linear model
    - Dealing with missing data
    - Categorical features
5. Bootstrap and Bagging
6. Random Forest

Thanks for attention!

Questions?

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